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## □ DSA Repository Progress Assessment

**Repo analyzed:** DS/ (11 folders, 26 PHP files, ~15,000 lines of solved code) **Method:** Every file was read/grepped for actual function implementations (not just comment headers) and cross-checked against each file’s own “Practice Problems & Applications” wish-lists, so the analysis reflects what is *genuinely coded*, not just what’s mentioned. All 28 standard interview patterns (the same taxonomy used in your DSA Patterns roadmap) are used as the reference framework. **Target bar:** Senior/Staff SDE — Saudi Arabia product companies, UAE (Dubai/Abu Dhabi) product companies, India Tier-1/Tier-2 (₹60LPA+).

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## □ Completed Patterns

These patterns have multiple real, working, well-annotated implementations (not just references) — good interview depth.

| Pattern                         | Evidence in Repo                                                                                                                                                                                                                                                                                 |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Two Pointers</b>             | Remove Duplicates (LC26), Move Zeroes (LC283), Sort Colors/Dutch Flag (LC75), Two Sum, 3Sum (LC15), 4Sum (LC18), Next Permutation (LC31), DLL pair-sum, Valid Palindrome (LC125)                                                                                                                 |
| <b>Sliding Window</b>           | Longest Substring Without Repeat (LC3, 2 approaches), Max Consecutive Ones III (LC1004, 2 approaches), Fruit Into Baskets (LC904, 2 approaches), Longest Repeating Char Replacement (LC424, 2 approaches), Binary Subarrays With Sum (LC930, atMost(K) trick), Number of Nice Subarrays (LC1248) |
| <b>Fast &amp; Slow Pointers</b> | Middle of LL (LC876), Cycle Detection (LC141, Floyd’s), Cycle Start (LC142, with math proof), Palindrome LL (LC234)                                                                                                                                                                              |
| <b>In-Place LL Reversal</b>     | Reverse Iterative + Recursive (LC206), Sort List via merge sort (LC148), Odd-Even LL (LC328)                                                                                                                                                                                                     |

| Pattern                               | Evidence in Repo                                                                                                                                                                                                                                                                          |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Binary Search (+ BS-on-Answer)</b> | First/Last Position (LC34), Search Rotated (LC33), Min in Rotated (LC153), Single Element (LC540), <b>Ship Packages (LC1011)</b> , <b>Aggressive Cows/Magnetic Force (LC1552)</b> , <b>Book Allocation</b> , <b>Split Array Largest Sum (LC410)</b> — the full BS-on-Answer canon is done |
| <b>Hashing</b>                        | Two Sum (LC1), Group Anagrams (LC49), Longest Consecutive Sequence (LC128), Majority Element I & II (LC169/229), Frequency of Most Frequent Element (LC1838)                                                                                                                              |
| <b>Monotonic Queue/Deque</b>          | Sliding Window Maximum (LC239) — the flagship problem for this pattern, fully implemented                                                                                                                                                                                                 |
| <b>Bit Manipulation</b>               | This is your <b>deepest</b> pattern: bit get/set/clear/toggle, XOR swap, $n \& (n-1)$ , power-of-2, Kernighan's bit count, Single Number (LC136) + Single Number III variant, subset generation via bitmask (LC78), divide without $*$ / $/$ (LC29), XOR range tricks                     |

## □ Partially Covered Patterns

Real code exists, but it stops short of the problems that actually get asked in senior-level interviews.

| Pattern                          | What's Done                                                                                                             | What's Missing                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sorting &amp; Cyclic Sort</b> | All 5 classical sorts (selection/bubble/insertion/merge/quicksort) implemented + dry-run; Sort Colors done in Array.php | The actual <b>Cyclic Sort placement technique</b> (index $\leftrightarrow$ value swapping) is never used — Missing Number was solved via XOR/sum instead. <b>First Missing Positive (LC41)</b> , Find All Duplicates (LC442), Find the Duplicate Number (LC287) are all absent, and these are extremely common at Amazon/Flipkart-tier bars |
| <b>Prefix Sum</b>                | Subarray Sum Equals K (LC560), Longest Subarray Sum=K, XOR-prefix tricks ( $1 \rightarrow N$ , $L \rightarrow R$ )      | True <b>Difference Array</b> range-update technique (Car Pooling LC1094, Corporate Flight Bookings LC1109) is completely absent                                                                                                                                                                                                             |

| Pattern                             | What's Done                                                                                                                     | What's Missing                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Monotonic Stack</b>              | Next Greater Element I (LC496), Valid Parentheses (LC20), Sort a Stack                                                          | The problems that actually distinguish strong monotonic-stack candidates are all unimplemented: <b>Daily Temperatures (LC739), NGE II/Circular (LC503), Largest Rectangle in Histogram (LC84), Trapping Rain Water — stack version (LC42), Remove K Digits (LC402), Asteroid Collision (LC735)</b>                                                                          |
| <b>Recursion &amp; Backtracking</b> | Fast Power (LC50), pick/not-pick subsequence generation, Combination Sum (LC39), Subset Sums, Subsets II with duplicates (LC90) | The <b>core backtracking canon is missing entirely: Permutations (LC46), N-Queens (LC51), Word Search (LC79), Palindrome Partitioning (LC131), Sudoku Solver, Generate Parentheses, Letter Combinations of a Phone Number.</b> This is a real gap — subset/combination groundwork is solid, but the “explore a tree of choices on a grid/board” skill hasn't been exercised |

## □ Missing Patterns

Zero implementations found anywhere in the repo (no folder, no function, no partial attempt) — these represent the majority of what a Senior SDE round actually tests.

| Pattern                       | Interview Relevance                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Merge Intervals</b>        | Very high (Amazon/logistics-heavy Gulf companies love scheduling/interval problems)                 |
| <b>Tree DFS</b>               | Extremely high — foundational for everything downstream                                             |
| <b>Tree BFS (Level Order)</b> | Extremely high                                                                                      |
| <b>Binary Search Trees</b>    | High                                                                                                |
| <b>Trie</b>                   | Medium (relevant for search/autocomplete features at Noon, Talabat, Zomato-style product companies) |
| <b>Heap / Priority Queue</b>  | Extremely high — Top-K, scheduling, streaming-median problems appear constantly                     |
| <b>Graph BFS/DFS</b>          | Extremely high — single largest source of Amazon/Google-style questions                             |

| Pattern                                      | Interview Relevance                                                                         |
|----------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Union-Find (DSU)</b>                      | High — connectivity/network problems, common at Google/Amazon                               |
| <b>Topological Sort</b>                      | High — course schedule/build-order style questions                                          |
| <b>Shortest Path (Dijkstra/Bellman-Ford)</b> | Medium-High (more common at Google/Staff-level bars)                                        |
| <b>DP - 1D</b>                               | Extremely high                                                                              |
| <b>DP - 2D/Grid</b>                          | Extremely high                                                                              |
| <b>DP - Knapsack</b>                         | High                                                                                        |
| <b>DP - Strings</b>                          | High (LCS/Edit Distance are near-universal)                                                 |
| <b>DP - Bitmask</b>                          | Medium (Staff-level/Google bar)                                                             |
| <b>Greedy</b>                                | High — interval scheduling, Gas Station, Task Scheduler style questions common in OA rounds |

**Bottom line on coverage:** you’ve built excellent depth across roughly the first third of the standard pattern taxonomy (array/string/LL/binary-search/bit-level thinking) but have not yet touched trees, graphs, heaps, DP, or greedy — which together account for the majority of medium/hard questions at the companies you’re targeting.

## □ Strengths

**Bit Manipulation mastery.** This is genuinely your strongest pattern — you’ve implemented get/set/clear/toggle bit, XOR swap, Kernighan’s algorithm, power-of-2 checks, bitmask subset generation, and arithmetic-without-operators tricks (divide, square). Very few candidates at any level go this deep voluntarily.

**Binary-Search-on-Answer fluency.** You’ve correctly implemented all four canonical “minimize the maximum / maximize the minimum” problems (Ship Packages, Aggressive Cows, Book Allocation, Split Array Largest Sum) — this is a notoriously hard-to-internalize meta-pattern, and having it solid puts you ahead of most candidates who only know textbook binary search.

**Multiple-approach discipline.** Across nearly every file, you write a brute-force version, then an optimized version, and often a third alternative — with dry runs and TC/SC annotations. This is exactly how strong candidates think out loud in interviews, and it shows in your Sliding Window and String files especially (e.g., `atMost(K)` trick reused across 3 different problems).

**Linked List manipulation.** Reversal (iterative + recursive), Floyd’s cycle detection with the mathematical proof of the cycle-start formula, and merge-sort-on-a-linked-list are all correctly implemented — this is comfortably interview-ready.

**Array + Hashing fundamentals.** Kadane’s algorithm, Boyer-Moore voting, Dutch National Flag, prefix-sum+hashmap for subarray-sum problems, and the N-Sum family (2/3/4Sum) are all done cleanly.

## □ Weak Areas

**No tree or graph thinking yet.** Zero DFS/BFS-on-a-tree-or-graph code exists. Since so many later patterns (Trie, Union-Find, Topological Sort, most DP-on-graphs) build on this mental model, this is the single highest-leverage gap to close.

**No DP exposure beyond Fibonacci-with-memoization.** DP is typically 20-30% of a Senior-level onsite loop across Amazon, Google, and most India Tier-1 companies. Right now you have no practiced intuition for state definition, transition, or base cases beyond the 1D Fibonacci seed.

**Backtracking stops at subsets/combinations.** You can generate subsets and combination sums, but you haven't yet practiced the "place a choice, recurse, undo" loop on a 2D board or fixed-size arrangement (N-Queens, Word Search, Sudoku, Permutations) — this is a different flavor of backtracking that shows up constantly in "hard" interview rounds.

**Merge Intervals is a complete blind spot** despite being one of the most frequently asked patterns at logistics/marketplace companies (highly relevant for Careem, Talabat, Noon, Amazon).

**"Hard" variants within your own strong patterns are being skipped.** You've solved the medium version of monotonic-stack, sliding-window, and array problems, but consistently stopped before the hard tier in the same category: Minimum Window Substring, Trapping Rain Water, Largest Rectangle in Histogram, Median of Two Sorted Arrays. These specific problems come up disproportionately often precisely because they're the hard-tier "filter" question in each pattern.

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## □ Recommended Next Patterns (Priority Order)

This order is optimized specifically for Saudi Arabia, UAE, and India Tier-1/Tier-2 (₹60LPA+) product-company loops, which lean heavily on **graphs, trees, heaps, DP, and greedy/interval scheduling** (the last one especially relevant given how many Gulf product companies are logistics/marketplace businesses — Careem, Talabat, Noon, Amazon).

1. **Merge Intervals** — quick win; you already have sorting mastered, this is a short, high-frequency pattern to close out first.
2. **Recursion & Backtracking (finish it)** — Permutations, N-Queens, Word Search, Palindrome Partitioning, Sudoku Solver. Natural extension of what you've already built.
3. **Tree DFS + Tree BFS** — unlocks everything downstream; do these together since most tree problems need both traversal styles.
4. **Binary Search Trees** — short pattern, builds directly on Tree DFS.
5. **Heap / Priority Queue** — Top-K, K-way merge, scheduling problems; extremely common at Amazon/Uber-style companies.
6. **Graph BFS/DFS** — the single highest-volume pattern in Gulf and India interview loops (grid problems, connectivity, multi-source BFS).
7. **Topological Sort + Union-Find** — natural graph extensions, frequently paired together in course-schedule/network-connectivity questions.
8. **DP - 1D → DP - 2D/Grid → DP - Knapsack → DP - Strings** — biggest total volume of medium/hard questions across every company on your target list; do them in this order since each builds intuition for the next.
9. **Greedy** — pairs naturally with your existing sorting/interval skills once Merge Intervals is done; high value for OA rounds.
10. **Trie, Shortest Path, DP - Bitmask** — lower priority unless you're specifically targeting Google or a Staff-level bar; do these last.

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## □ Practice Plan (Problems per Pattern)

| Pattern                      | Est. Problems to Reach Interview-Ready | Why It Is Important                                                             | Difficulty Curve | Top Problems (Easy → Medium → Hard)                                                                                                                    |
|------------------------------|----------------------------------------|---------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Merge Intervals</b>       | 8-10                                   | Very common scheduling/logistics question at Gulf marketplace companies         | Easy → Medium    | Merge Intervals (56) → Insert Interval (57) → Non-overlapping Intervals (435) → Meeting Rooms II (253, premium)                                        |
| <b>Backtracking (finish)</b> | 12-15                                  | “Explore all choices” is a top-3 tested skill across every company on your list | Easy → Hard      | Subsets (78) → Permutations (46) → Combination Sum II (40) → Word Search (79) → N-Queens (51) → Sudoku Solver (37)                                     |
| <b>Tree DFS</b>              | 15-18                                  | Foundational; unlocks BST/Trie/Graph reasoning                                  | Easy → Hard      | Max Depth of Binary Tree (104) → Diameter of Binary Tree (543) → Path Sum II (113) → Lowest Common Ancestor (236) → Binary Tree Maximum Path Sum (124) |
| <b>Tree BFS</b>              | 8-10                                   | Level-order thinking, common warm-up round question                             | Easy → Medium    | Binary Tree Level Order (102) → Right Side View (199) → Zigzag Level Order (103) → Minimum Depth (111)                                                 |
| <b>Binary Search Trees</b>   | 8-10                                   | Property-based tree reasoning, common at Microsoft/Amazon                       | Easy → Hard      | Validate BST (98) → Kth Smallest in BST (230) → Insert into BST (701) → LCA of BST (235) → Recover BST (99)                                            |

| Pattern                      | Est. Problems to Reach Interview-Ready | Why It Is Important                                            | Difficulty Curve | Top Problems (Easy → Medium → Hard)                                                                                                               |
|------------------------------|----------------------------------------|----------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Heap / Priority Queue</b> | 12-15                                  | Top-K and scheduling problems appear constantly at Amazon/Uber | Easy → Hard      | Kth Largest Element (215) → Top K Frequent Elements (347) → Task Scheduler (621) → Merge K Sorted Lists (23) → Find Median from Data Stream (295) |
| <b>Graph BFS/DFS</b>         | 18-20                                  | Highest-volume pattern for your target companies               | Easy → Hard      | Number of Islands (200) → Rotting Oranges (994) → Clone Graph (133) → Course Schedule (207) → Word Ladder (127)                                   |
| <b>Union-Find</b>            | 6-8                                    | Connectivity problems, common at Google/Amazon                 | Medium → Hard    | Number of Provinces (547) → Redundant Connection (684) → Accounts Merge (721) → Number of Islands II (305, premium)                               |
| <b>Topological Sort</b>      | 6-8                                    | Build-order/dependency questions                               | Medium           | Course Schedule (207) → Course Schedule II (210) → Alien Dictionary (269, premium)                                                                |
| <b>DP - 1D</b>               | 12-15                                  | Foundational DP intuition                                      | Easy → Medium    | Climbing Stairs (70) → House Robber (198) → Coin Change (322) → Longest Increasing Subsequence (300)                                              |
| <b>DP - 2D/Grid</b>          | 10-12                                  | Very common at Google/Flipkart-tier                            | Medium → Hard    | Unique Paths (62) → Minimum Path Sum (64) → Dungeon Game (174)                                                                                    |

| Pattern              | Est. Problems to Reach Interview-Ready | Why It Is Important                                  | Difficulty Curve | Top Problems (Easy → Medium → Hard)                                                                                         |
|----------------------|----------------------------------------|------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>DP - Knapsack</b> | 8-10                                   | Resource-allocation style questions                  | Medium           | Partition Equal Subset Sum (416) → Target Sum (494) → Coin Change II (518)                                                  |
| <b>DP - Strings</b>  | 8-10                                   | LCS/Edit Distance are near-universal at Senior bar   | Medium → Hard    | Longest Common Subsequence (1143) → Edit Distance (72) → Palindromic Substrings DP (647) → Regular Expression Matching (10) |
| <b>Greedy</b>        | 8-10                                   | OA-round staple, pairs with Merge Intervals          | Easy → Hard      | Assign Cookies (455) → Jump Game (55) → Gas Station (134) → Task Scheduler (621)                                            |
| <b>Trie</b>          | 5-6                                    | Lower priority unless search/autocomplete-heavy role | Medium           | Implement Trie (208) → Word Search II (212)                                                                                 |
| <b>Shortest Path</b> | 5-6                                    | Staff/Google-tier bar                                | Medium → Hard    | Network Delay Time (743) → Cheapest Flights Within K Stops (787)                                                            |

**Also finish inside patterns you already “completed”:** Minimum Window Substring (76), Trapping Rain Water (42), Largest Rectangle in Histogram (84), Median of Two Sorted Arrays (4), Daily Temperatures (739), First Missing Positive (41) — these hard-tier problems within your strong patterns are disproportionately likely to appear as the “filter” question in a Senior-level loop.

## □ Estimated Interview Readiness (%)

≈ 35% overall, weighted for a Senior/Staff bar.

Reasoning: using the 28-pattern taxonomy as the base, roughly 8 patterns are solidly done, 4 are half-done, and 16 are untouched — a raw weighted score of  $(8 \times 1.0 + 4 \times 0.5 + 16 \times 0) / 28 = 36\%$ . This aligns with a qualitative read too: your **array, string, linked-list, binary-search, and bit-manipulation fundamentals are genuinely strong (85-90% depth within those patterns)** — but since trees, graphs, heaps, DP, and greedy collectively make up the majority of medium/hard questions in Senior-level loops at your target companies, overall readiness sits closer to a third of the way there. This is a narrow-but-deep profile: you will pass early/easy rounds comfortably, but need the missing 16 patterns to clear a full Senior-level loop (typically 4-5 rounds, at least 2-3 of which will hit trees/graphs/DP/heap directly).



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## □ Final Roadmap to Completion

Assuming ~10-12 hours/week of practice, here's a realistic sequencing to interview-ready:

| Phase                                                            | Weeks | Focus                                                                                                         | Patterns Covered                                                                                                               |
|------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 1 — Close the Gaps</b>                                  | 1-2   | Finish what's already started                                                                                 | Merge Intervals, Backtracking (Permutations/N-Queens/Word Search/Sudoku), hard-tier problems in Sliding Window/Monotonic Stack |
| <b>Phase 2 — Tree Foundations</b>                                | 3-4   | Build the tree mental model                                                                                   | Tree DFS, Tree BFS, Binary Search Trees                                                                                        |
| <b>Phase 3 — Heap &amp; Graph Core</b>                           | 5-7   | Highest-volume patterns for your target companies                                                             | Heap/Priority Queue, Graph BFS/DFS, Union-Find, Topological Sort                                                               |
| <b>Phase 4 — Dynamic Programming</b>                             | 8-10  | Largest single point-scoring category                                                                         | DP-1D → DP-2D/Grid → DP-Knapsack → DP-Strings                                                                                  |
| <b>Phase 5 — Greedy &amp; Polish</b>                             | 11    | Round out remaining common patterns                                                                           | Greedy, Trie                                                                                                                   |
| <b>Phase 6 — Advanced/Staff Polish</b>                           | 12    | Only if targeting Google or Staff-level                                                                       | Shortest Path, DP-Bitmask                                                                                                      |
| <b>Phase 7 — Mock Interviews &amp; Company-Specific Drilling</b> | 13-14 | Timed mocks, company tag lists (Amazon/Google/Flipkart tags on LeetCode), system-design-lite for Senior loops | Cross-pattern integration                                                                                                      |

By the end of **Phase 4 (~Week 10)**, you'll have covered every pattern that appears in the large majority of Saudi/UAE/India Tier-1 Senior-level loops, putting realistic readiness at **75-80%**. Phases 5-7 close the remaining gap to a genuinely comprehensive, Staff-adjacent preparation level.